

B. Anechoic Wind Tunnel and Acoustic Instrumentation

The NASA Glenn 9- by 15-Foot LSWT is located in the low speed return leg of the 8- by 6-Foot Supersonic Wind Tunnel. The tunnel test section walls, floor, and ceiling have acoustic treatment to produce an anechoic test environment. Figure 5 is a sketch of the research fan stage installed in the 9- by 15-Foot LSWT. Sideline acoustic data were acquired with a computer-controlled translating microphone probe and with three aft microphone assemblies mounted to the tunnel floor. The translating microphone probe acquired data at 48 sideline geometric angles from 27.2 to 134.6° relative to the rotor plane. The translating probe traverse was 89 in. from the rotor rotational axis (about four rotor diameters). Acoustic data taken at this sideline distance are representative of far-field conditions, although there are still manifestations of near field inlet/exhaust noise lobes. A wall-mounted microphone probe was placed at a reference location adjacent to the translating probe home position (134.6° , maximum aft travel). The three fixed microphone assemblies were mounted at the home axial position to acquire aft acoustic data at geometric angles of 140 , 150 , and 160° . Data were also acquired with an acoustic barrier wall installed adjacent to the fan stage which effectively blocked aft-radiated noise, allowing for measurement of only the inlet-radiated noise (fig. 6). The acoustic data were acquired through a digital computer system and stored for post-run analysis.

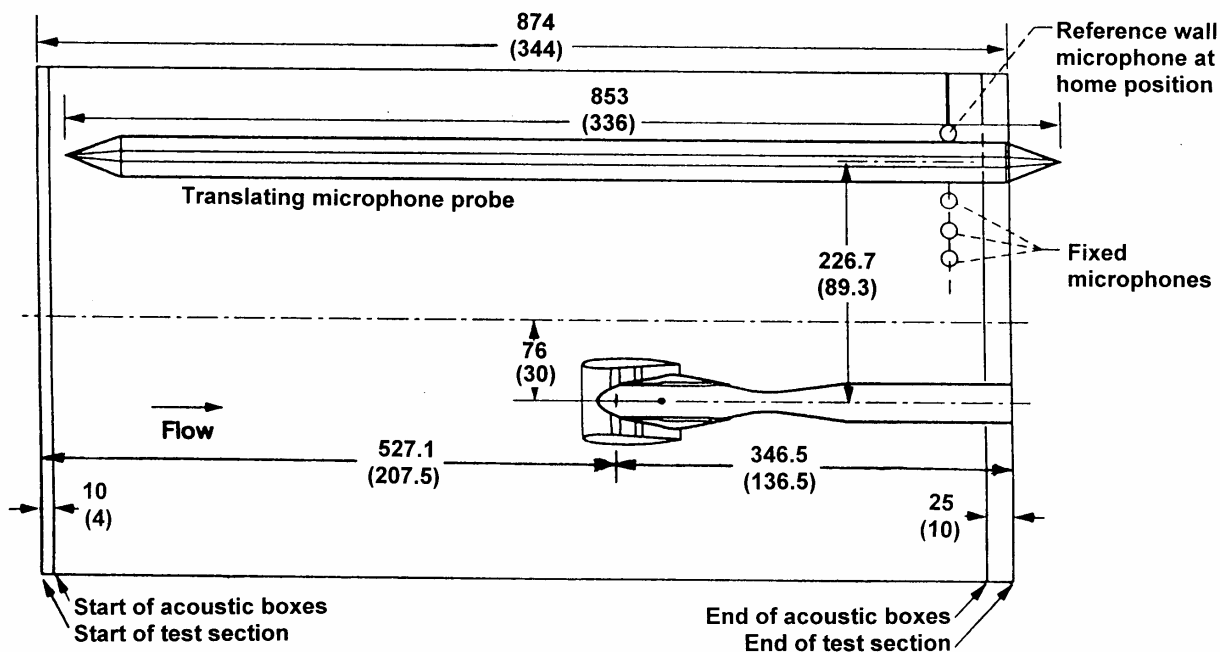


Figure 5.—Sketch of the model fan stage installed in the 9- by 15- Foot LSWT. Far-field acoustic data were acquired with a translating microphone probe and aft fixed microphones (dimensions in cm (in.)).

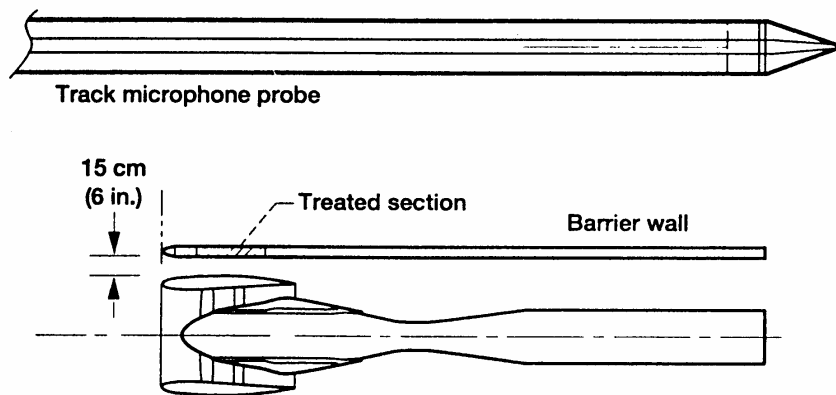


Figure 6.—Sketch showing location of acoustic barrier wall relative to model fan stage (dimensions in cm (in.)).